SIM1003 Calculus II
Tutorial 2

All even-numbered questions in the following exercises.

EXERCISE 8.2 Integration by Parts

Integration by Parts

Evaluate the integrals in Exercises 1 – 24 using integrals by parts.

1. $\int x \sin \frac{x}{2}\, dx$
2. $\int \theta \cos \pi\theta\, d\theta$
3. $\int t^2 \cos\, t\, dt$
4. $\int x^2 \sin\, x\, dx$
5. $\int_1^2 x \ln x\, dx$
6. $\int_1^e x^3 \ln x\, dx$
7. $\int xe^x\, dx$
8. $\int xe^{3x}\, dx$
9. $\int x^2 e^{-x}\, dx$
10. $\int (x^2 - 2x + 1)e^{2x}\, dx$
11. $\int \tan^{-1} y\, dy$
12. $\int \arcsin y\, dy$
13. $\int x\, \sec^2 x\, dx$
14. $\int 4x\, \sec^2 2x\, dx$
15. $\int x^3 e^x\, dx$
16. $\int p^4 e^{-p}\, dp$
17. $\int (x^2 - 5x)e^x\, dx$
18. $\int (r^2 + r + 1)e^r\, dr$
19. $\int x^5 e^x\, dx$
20. $\int t^2 e^{4t}\, dt$
21. $\int e^\theta\, \sin\, \theta\, d\theta$
22. $\int e^{-y} \cos\, y\, dy$
23. $\int e^{2x} \cos\, 3x\, dx$
24. $\int e^{-2x} \sin\, 2x\, dx$

Using Substitutions

Evaluate the integrals in Exercises 25 – 30 by using a substitution prior to integration by parts.

25. $\int e^{\sqrt{3s+9}}\, ds$

26. $\int_0^1 x\sqrt{1-x}\, dx$

27. $\int_0^{\pi/3} x\, \tan^2 x\, dx$

28. $\int \ln\left(x + x^2\right)\, dx$

29. $\int \sin(\ln x)\, dx$

30. $\int z(\ln z)^2\, dz$

Evaluating Integrals

Evaluate the integrals in Exercises 31 – 56. Some integrals do not require integration by parts.

31. $\int x\, \sec\, x^2\, dx$

32. $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\, dx$

33. $\int x\left(\ln x\right)^2\, dx$

34. $\int \frac{1}{x(\ln x)^2}\, dx$

35. $\int \frac{\ln x}{x^2}\, dx$

36. $\int \frac{(\ln x)^3}{x}\, dx$

37. $\int x^3\, e^{x^4}\, dx$

38. $\int x^5\, e^{x^3}\, dx$

39. $\int x^3\sqrt{x^2+1}\, dx$

40. $\int x^2\, \sin\, x^3\, dx$

41. $\int \sin\, 3x\, \cos 2x\, dx$

42. $\int \sin 2x \cos 4x\, dx$

43. $\int \sqrt{x}\, \ln x\, dx$

44. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\, dx$

45. $\int \cos\, \sqrt{x}\, dx$

46. $\int \sqrt{x}e^{\sqrt{x}}\, dx$

47. $\int_0^{\pi/2} \theta^2\, \sin\, 2\theta\, d\theta$

48. $\int_0^{\pi/2} x^3\, \cos\, 2x\, dx$

49. $\int_{2/\sqrt{3}}^{2} t\sec^{-1}\, t\, dt$

50. $\int_0^{1/\sqrt{2}} 2x\, \arcsin\left(x^2\right)\, dx$

51. $\int x\, \arctan\, x\, dx$

52. $\int x^2\, \tan^{-1}\, \frac{x}{2} dx$

53. $\int \left(1+2x^2\right)e^{x^2}\, dx$

54. $\int \frac{xe^x}{(x+1)^2}\, dx$

55. $\int \sqrt{x}\left(\arcsin\sqrt{x}\right)\, dx$

56. $\int \frac{\left(\sin^{-1} x\right)^2}{\sqrt{1-x^2}}\, dx$

EXERCISE 8.3 Trigonometric Integrals

Power of Sines and Cosines

Evaluate the integrals in Exercises 1 – 22.

1. $\int \cos\ 2x\ dx$
2. $\int_0^{\pi} 3 \sin \frac{x}{3}\ dx$
3. $\int \cos^3\ x\ \sin\ x\ dx$
4. $\int \sin^4\ 2x\ \cos\ 2x\ dx$
5. $\int \sin^3 x\ dx$
6. $\int \cos^3\ 4x\ dx$
7. $\int \sin^5 x\ dx$
8. $\int_0^{\pi} \sin^5 \frac{x}{2}\ dx$
9. $\int \cos^3\ x\ dx$
10. $\int_0^{\pi/6} 3\ \cos^5\ 3x\ dx$
11. $\int \sin^3\ x\ \cos^3\ x\ dx$
12. $\int \cos^3\ 2x\ \sin^5\ 2x\ dx$
13. $\int \cos^2 x\ dx$
14. $\int_0^{\pi/2} \sin^2 x\ dx$
15. $\int_0^{\pi/2} \sin^7 y\ dy$
16. $\int 7 \cos^7 t\ dt$
17. $\int_0^{\pi} 8 \sin^4 x\ dx$
18. $\int 8 \cos^4\ 2\pi x\ dx$
19. $\int 16\ \sin^2 x\ \cos^2 x\ dx$
20. $\int_0^{\pi} 8\ \sin^4 y\ \cos^2 y\ dy$
21. $\int 8\ \cos^3\ 2\theta\ \sin\ 2\theta\ d\theta$
22. $\int_0^{\pi/2} \sin^2\ 2\theta\ \cos^3\ 2\theta\ d\theta$

Integrating Square roots

Evaluate the integrals in Exercises 23 - 32.

23. $\int_0^{2\pi} \sqrt{\frac{1-\cos x}{2}}\ dx$
24. $\int_0^{\pi} \sqrt{1-\cos\ 2x}\ dx$
25. $\int_0^{\pi} \sqrt{1-\sin^2\ t}\ dt$
26. $\int_0^{\pi} \sqrt{1-\cos^2\ \theta}\ d\theta$
27. $\int_{\pi/3}^{\pi/2} \frac{\sin^2 x}{\sqrt{1-\cos x}}\ dx$
28. $\int_0^{\pi/6} \sqrt{1+\sin\ x}\ dx$

    ($Hint$: Multiply by $\sqrt{\frac{1-\sin x}{1-\sin x}}$.)
29. $\int_{5\pi/6}^{\pi} \frac{\cos^4\ x}{\sqrt{1-\sin\ x}}\ dx$
30. $\int_{\pi/2}^{3\pi/4} \sqrt{1-\sin\ 2x}\ dx$
31. $\int_0^{\pi/2} \theta\sqrt{1-\cos\ 2\theta}\ d\theta$
32. $\int_{-\pi}^{\pi} \left(1-\cos^2 t\right)^{3/2}\ dt$

Power of Tangents and Secants

Evaluate the integrals in Exercises 33 - 52.

33. $\int \sec^2 x \tan x \, dx$

34. $\int \sec x \tan^2 x \, dx$

35. $\int \sec^3 x \tan x \, dx$

36. $\int \sec^3 x \tan^3 x \, dx$

37. $\int \sec^2 x \tan^2 x \, dx$

38. $\int \sec^4 x \tan^2 x \, dx$

39. $\int_{-\pi/3}^{0} 2 \sec^3 x \, dx$

40. $\int e^x \sec^3 e^x \, dx$

41. $\int \sec^4 \theta \, d\theta$

42. $\int \tan^4 x \sec^3 x \, dx$

43. $\int_{\pi/4}^{\pi/2} \csc^4 \theta \, d\theta$

44. $\int \sec^6 x \, dx$

45. $\int 4 \tan^3 x \, dx$

46. $\int_{-\pi/4}^{\pi/4} 6 \tan^4 x \, dx$

47. $\int \tan^5 x \, dx$

48. $\int \cot^6 2x \, dx$

49. $\int_{\pi/6}^{\pi/3} \cot^3 x \, dx$

50. $\int 8 \cot^4 t \, dt$

51. $\int_{\pi/4}^{\pi/3} \tan^5 \theta \sec^4 \theta \, d\theta$

52. $\int \cot^3 t \csc^4 t \, dt$

Products of Sines and Cosines

Evaluate the integrals in Exercises 53 - 58.

53. $\int \sin 3x \cos 2x \, dx$

54. $\int \sin 2x \cos 3x \, dx$

55. $\int_{-\pi}^{\pi} \sin 3x \sin 3x \, dx$

56. $\int_{0}^{\pi/2} \sin x \cos x \, dx$

57. $\int \cos 3x \cos 4x \, dx$

58. $\int_{-\pi/2}^{\pi/2} \cos x \cos 7x \, dx$

Using Identities

Evaluate the integrals in Exercises 59 - 64.

59. $\int \sin^2 \theta \cos 3\theta \, d\theta$

60. $\int \cos^2 2\theta \sin \theta \, d\theta$

61. $\int \cos^3 \theta \sin 2\theta \, d\theta$

62. $\int \sin^3 \theta \cos 2\theta \, d\theta$

63. $\int \sin \theta \cos \theta \cos 3\theta \, d\theta$

64. $\int \sin \theta \sin 2\theta \sin 3\theta \, d\theta$

EXERCISE 8.4 Trigonometric Substitutions

Using Trigonometric Substitutions

Evaluate the integrals in Exercises 1 - 14.

1. $\int \frac{dx}{\sqrt{9+x^2}}$

2. $\int \frac{3\,dx}{\sqrt{1+9x^2}}$

3. $\int_{-2}^{2} \frac{dx}{4+x^2}$

4. $\int_{0}^{2} \frac{dx}{8+2x^2}$

5. $\int_{0}^{3/2} \frac{dx}{\sqrt{9-x^2}}$

6. $\int_{0}^{1/2\sqrt{2}} \frac{2\,dx}{\sqrt{1-4x^2}}$

7. $\int \sqrt{25-t^2}\, dt$

8. $\int \sqrt{1-9t^2}\, dt$

9. $\int \frac{dx}{\sqrt{4x^2-49}}, \quad x > \frac{7}{2}$

10. $\int \frac{5\,dx}{\sqrt{25x^2-9}}, \quad x > \frac{3}{5}$

11. $\int \frac{\sqrt{y^2-49}}{y}\, dy, \quad y > 7$

12. $\int \frac{\sqrt{y^2-25}}{y^3}\, dy, \quad y > 5$

13. $\int \frac{dx}{x^2\sqrt{x^2-1}}, \quad x > 1$

14. $\int \frac{2dx}{x^3\sqrt{x^2-1}}, \quad x > 1$

Assorted Integrations

Use any method to evaluate the integrals in Exercises 15 - 52.

15. $\displaystyle\int \frac{dx}{x\sqrt{x^2-1}}$

16. $\displaystyle\int \frac{dx}{1+x^2}$

17. $\displaystyle\int \frac{x\,dx}{\sqrt{x^2-1}}$

18. $\displaystyle\int \frac{dx}{\sqrt{1-x^2}}$

19. $\displaystyle\int \frac{x}{\sqrt{9-x^2}}\,dx$

20. $\displaystyle\int \frac{x^2}{4+x^2}\,dx$

21. $\displaystyle\int \frac{x^3\,dx}{\sqrt{x^2+4}}$

22. $\displaystyle\int \frac{dx}{x^2\sqrt{x^2+1}}$

23. $\displaystyle\int \frac{8\,dw}{w^2\sqrt{4-w^2}}$

24. $\displaystyle\int \frac{\sqrt{9-w^2}}{w^2}\,dw$

25. $\displaystyle\int \sqrt{\frac{x+1}{1-x}}\,dx$

26. $\displaystyle\int x\sqrt{x^2-4}\,dx$

27. $\displaystyle\int_0^{\sqrt{3}/2} \frac{4x^2\,dx}{(1-x^2)^{3/2}}$

28. $\displaystyle\int_0^1 \frac{dx}{(4-x^2)^{3/2}}$

29. $\displaystyle\int \frac{dx}{(x^2-1)^{3/2}}, \quad x>1$

30. $\displaystyle\int \frac{x^2\,dx}{(x^2-1)^{5/2}}, \quad x>1$

31. $\displaystyle\int \frac{(1-x^2)^{3/2}}{x^6}\,dx$

32. $\displaystyle\int \frac{(1-x^2)^{1/2}}{x^4}\,dx$

33. $\displaystyle\int \frac{8\,dx}{(4x^2+1)^2}$

34. $\displaystyle\int \frac{6\,dt}{(9t^2+1)^2}$

35. $\displaystyle\int \frac{x^3\,dx}{x^2-1}$

36. $\displaystyle\int \frac{x\,dx}{25+4x^2}$

37. $\displaystyle\int \frac{v^2\,dv}{(1-v^2)^{5/2}}$

38. $\displaystyle\int \frac{(1-r^2)^{5/2}}{r^8}\,dr$

39. $\displaystyle\int_0^{\ln 4} \frac{e^t\,dt}{\sqrt{e^{2t}+9}}$

40. $\displaystyle\int_{\ln(3/4)}^{\ln(4/3)} \frac{e^t\,dt}{(1+e^{2t})^{3/2}}$

41. $\displaystyle\int_{1/12}^{1/4} \frac{2\,dt}{\sqrt{t}+4t\sqrt{t}}$

42. $\displaystyle\int_1^e \frac{dy}{y\sqrt{1+(\ln y)^2}}$

43. $\displaystyle\int \frac{x\,dx}{\sqrt{1+x^4}}$

44. $\displaystyle\int \frac{\sqrt{1-(\ln x)^2}}{x\ln x}\,dx$

45. $\displaystyle\int \sqrt{\frac{4-x}{x}}\,dx$
(*Hint*: Let $x=u^2$.)

46. $\displaystyle\int \sqrt{\frac{x}{1-x^3}}\,dx$
(*Hint*: Let $u=x^{3/2}$.)

47. $\displaystyle\int \sqrt{x}\sqrt{1-x}\,dx$

48. $\displaystyle\int \frac{\sqrt{x-2}}{\sqrt{x-1}}\,dx$

49. $\displaystyle\int \sqrt{8-2x-x^2}\,dx$

50. $\displaystyle\int \frac{1}{\sqrt{x^2-2x+5}}\,dx$

51. $\displaystyle\int \frac{\sqrt{x^2+4x+3}}{x+2}\,dx$

52. $\displaystyle\int \frac{\sqrt{x^2+2x+2}}{x^2+2x+1}\,dx$